VISTRAN: Need-Based Teacher Training Platform

Hackathon Submission – Theme 2: DIET & SCERT Teacher Training

Our Innovators: Team void 4

Om Chaudhari
omchaudhari0709@gmail.com
Contact: +91 87669 77328

Ayush Yadav
ayushyadav913012@gmail.com
Contact: +91 93224 45089

Siddhi Bahutule
siddhivilas26@gmail.com
Contact: +91 788 762 6385

Aisha Inamdar
inamdaraisha03@gmail.com
Contact: +91 9209264883

Executive Summary: The Challenge & Solution

The Challenge:

Dr. Kumar, a DIET principal, struggles with "training fatigue." Teachers find centralized, 50-page training manuals irrelevant to their specific classroom realities (Tribal vs. Urban). He lacks a real-time feedback loop to know what problems teachers are actually facing.

The Solution:

VISTRAN is a dynamic, AI-powered platform that replaces static manuals with a "Live Training Ecosystem." It allows teachers to report specific issues, uses AI to adapt training content to the local context (Language/Culture), and provides instant AI support—all within a unified Java-based architecture.

Our Approach: Demand-Driven Training

1

A. Problem-First Methodology

- Step 1: Teachers report immediate blockers (e.g., 'Student Absenteeism,' 'Hard Science Concepts') via the Issue Reporting System.
- Step 2: The system aggregates this data for Administrators (DIET Principals) to visualize cluster-wise trends.
- Step 3: Training is assigned in response to these specific data points.

2

B. Demographic Contextualization

Contextual AI: We use Generative AI not just to summarize, but to rewrite content. A standard science lesson is automatically simplified and culturally adapted for a tribal context using familiar local examples, ensuring higher engagement.

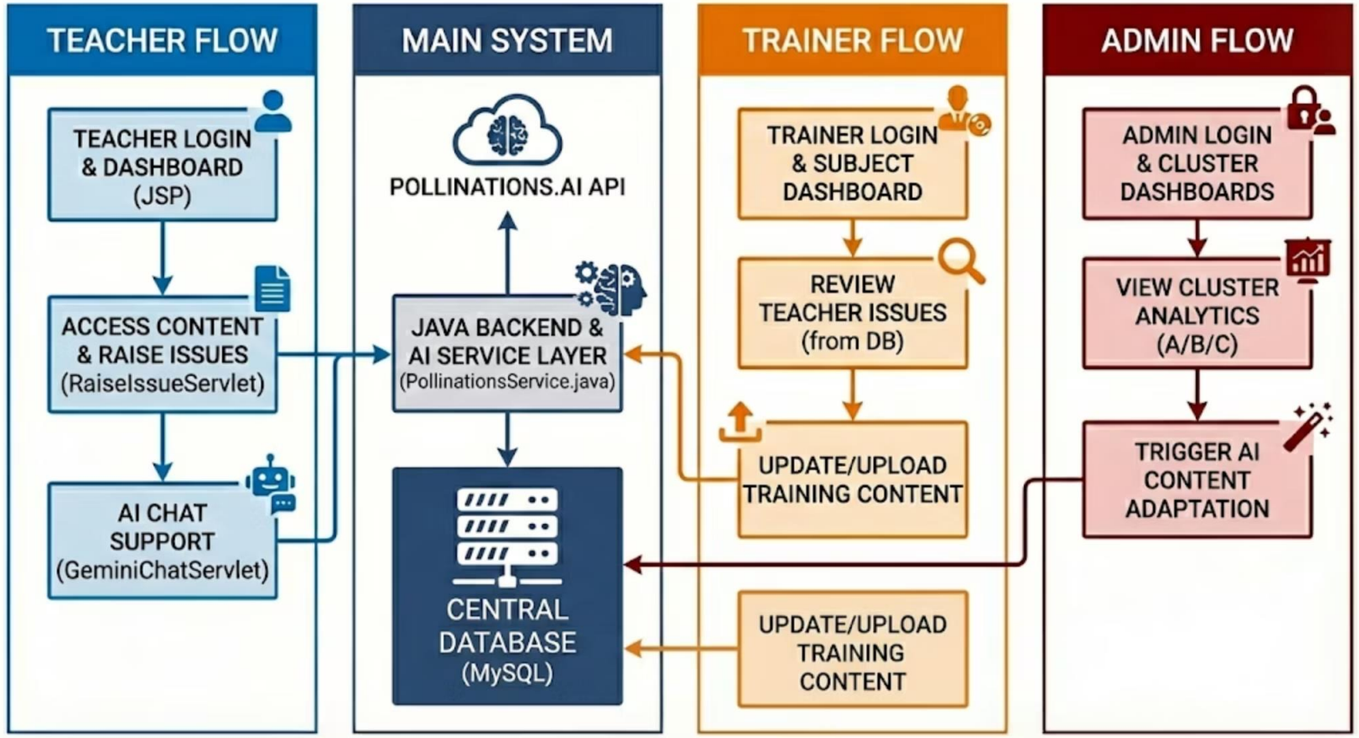
3

C. Role-Based Architecture

- Admin: High-level analytics and training deployment.
- Trainer: Content creation and issue resolution.
- Teacher: Learning access, issue raising, and AI assistance.

Vistran Architecture: A Holistic Approach

Our methodology ensures a robust, interconnected system for comprehensive teacher training and support.



Solution in Detail: Technical Implementation

A. Smart AI Content Adaptation (Java Service)

File: `src/main/java/com/user/PollinationsService.java`

This dedicated Java service intercepts standard training materials and processes them through the Pollinations.ai API to generate three distinct versions: Urban (Cluster A): Modern, practical language with high-resource examples. Semi-Urban (Cluster B): Balanced language with everyday Indian examples. Tribal (Cluster C): Simplified vocabulary, short sentences, and culturally familiar analogies.

B. Integrated AI Chat System (Servlet)

File: `src/main/java/com/user/GeminiChatServlet.java`

- **Replacement:** This Servlet completely replaces the previous Node.js server.
- **Function:** Handles all chatbot interactions directly within the Tomcat container. It acts as an intelligent 'Pedagogical Assistant,' helping teachers troubleshoot classroom management issues instantly without waiting for a human trainer.

C. Real-Time Feedback Loop

File: `src/main/java/com/user/RaiseIssueServlet.java`

- **Function:** Captures specific classroom issues tagged by Class and Subject.
- **Impact:** Reduces the feedback cycle from months (paper reports) to seconds (database entry), allowing DIETs to react immediately.

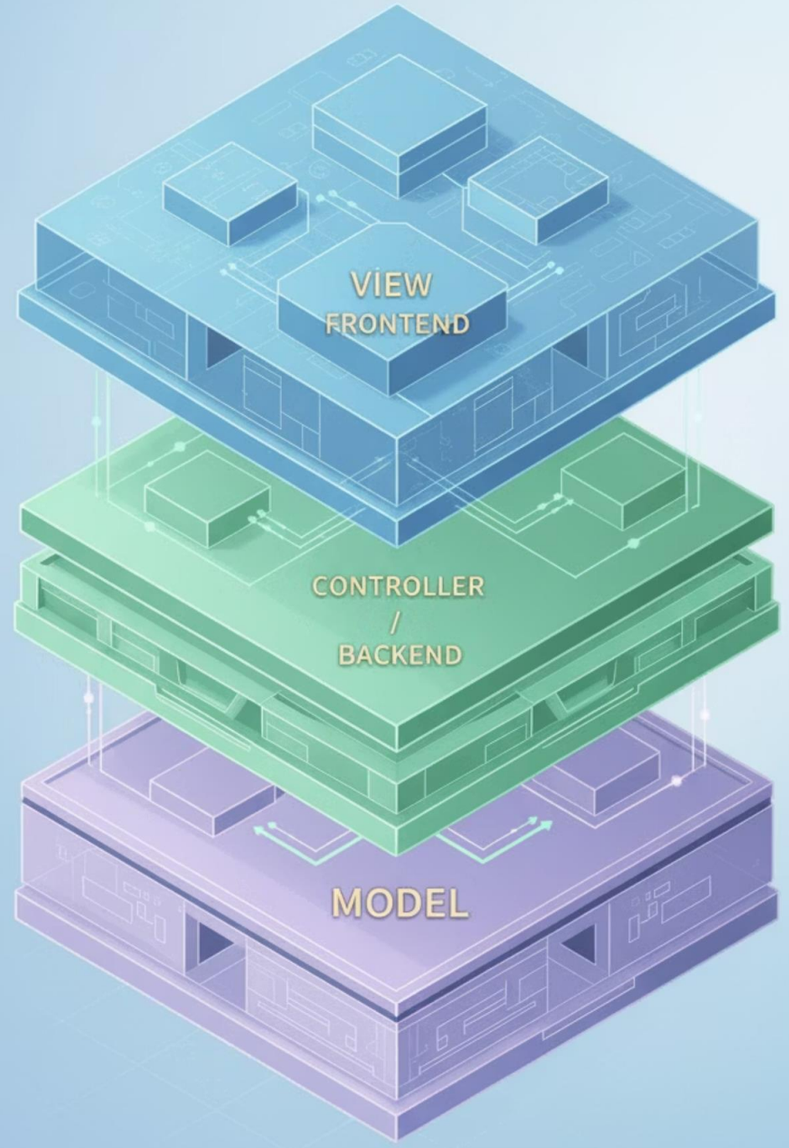
System Architecture: Java MVC Model

The system follows a pure Java MVC (Model-View-Controller) architecture:

- **View (Frontend):** JSP Pages (`clusterC.jsp`, `subjectdashboard.jsp`), HTML5, CSS3, Material Design.
- **Controller (Backend):** Jakarta EE Servlets manage data flow and AI logic.
  - `GeminiChatServlet`: AI Chatbot controller.
  - `AdaptModuleServlet`: Training content adaptation controller.
- **Model (Data):** MySQL Database accessed via `DBConnection.java` (Singleton Pattern).

Tech Stack:

- Frontend: JSP, HTML, CSS, JavaScript
- Backend: Java (Jakarta Servlets)
- Database: MySQL
- AI Service: Pollinations.ai
- Server: Apache Tomcat 10.1



Live Deployment & Access

Deployment Link: vistran.koyeb.app/

GitHub Repository: https://github.com/phantom550/Vistran

How to Run Locally:

Prerequisites:

- Java Development Kit (JDK): Version 21 or higher.
- Web Server: Apache Tomcat 10.1.
- Database: MySQL Server 8.0.
- IDE: Eclipse IDE (Enterprise Java edition).

Step 1: Database Configuration

1. Open MySQL Workbench.
2. Create the database: `CREATE DATABASE void4;`
3. Import the SQL schema provided in the `/database` folder of the repo.
4. Note: The app connects via `jdbc:mysql://localhost:3306/void4` (User: `root`, Pass: `root`) by default.

Step 2: Server Setup (No Node.js Required)

1. Open Eclipse IDE.
2. Import the project: File > Open Projects from File System > Select `'vistran by void4'`.
3. Right-click the project > Build Path > Configure Build Path.
4. Add Apache Tomcat v10.1 to Libraries.
5. Add `mysql-connector-j-9.5.0.jar` to Classpath.
6. Right-click Project > Run As > Run on Server.

Step 3: Access the Platform

- Home URL: http://localhost:8080/vistran_by_void4/
- Admin Login: Access via `adminlogin.jsp`.

Thank You!

Team void 4 - VISTRAN: Transforming Teacher Training in India